

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID: **SIH26167**

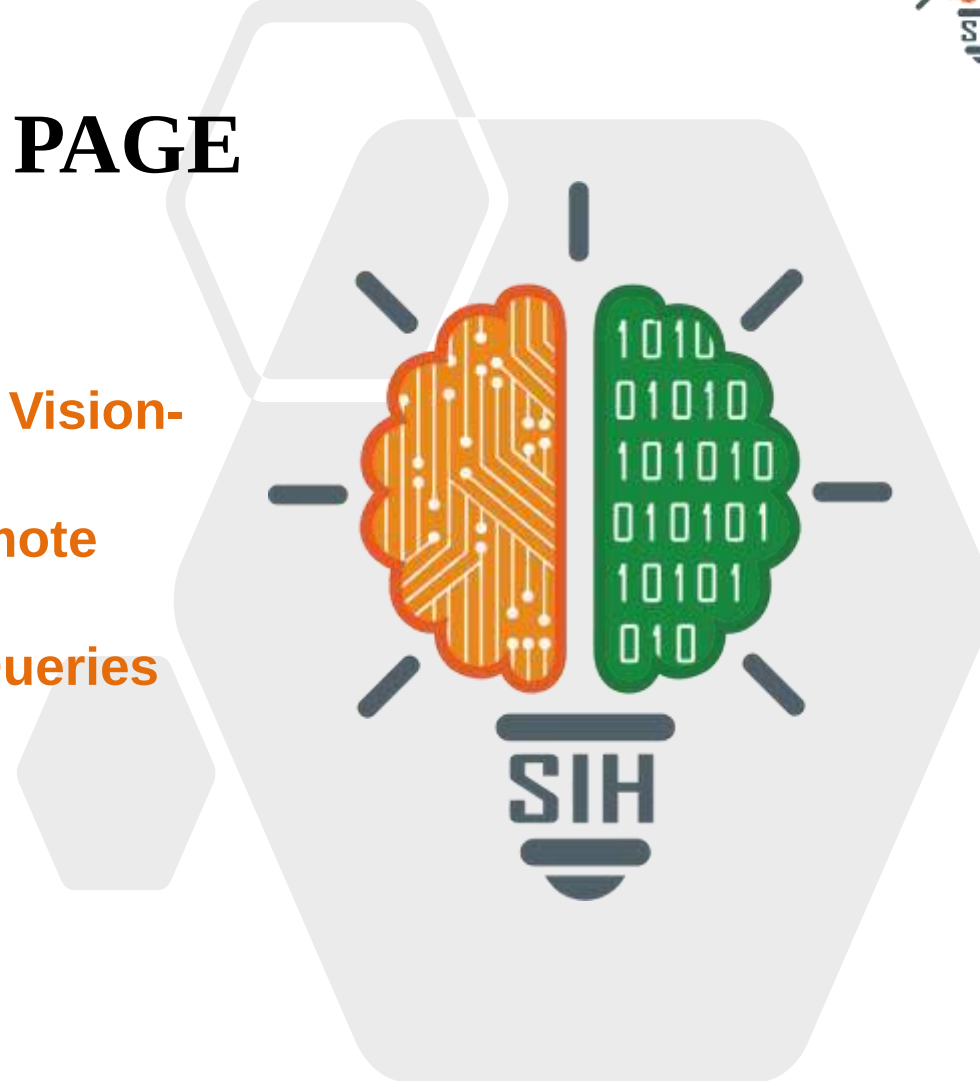
Problem Statement Title: **An Interactive Vision-
Language Assistant for Multimodal Remote
Sensing Image Analysis through Text Queries**

Theme: **Space Technology**

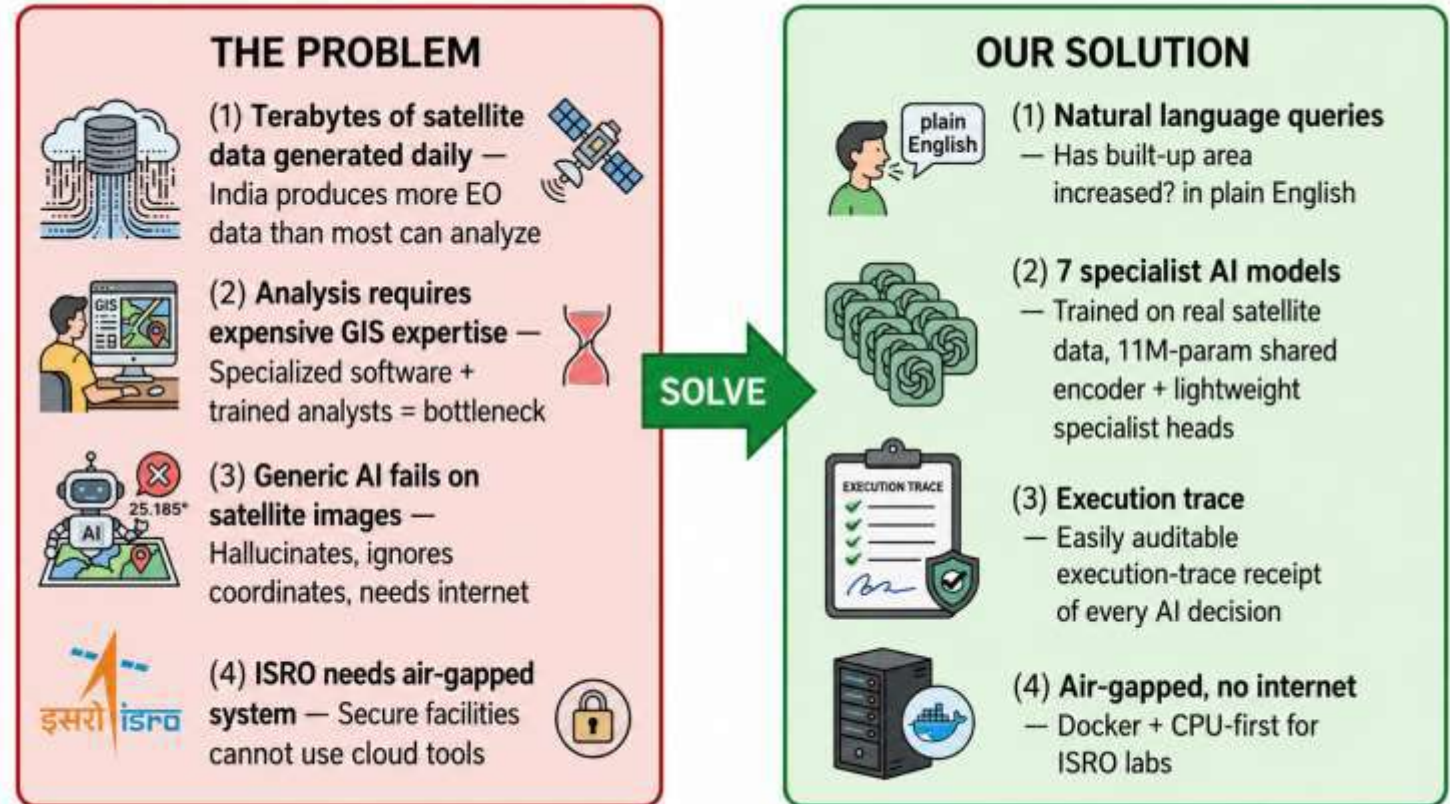
PS Category: **Software**

Team ID:

Team Name: **MCS**



- Satellites send down terabytes of imagery daily, but every question still needs a GIS expert, **expensive software**, and **days of manual analysis**
- Ask in **plain English**, **get evidence**, not text
Our agent routes every query to specialist models and returns answers **with** confidence scores, highlighted map overlays **not** paragraphs
- **Reliability:** Every answer comes with a trail
Each decision → task chosen, model used, parameters bound, time taken recorded in auditable execution trace, so an analyst can defend any finding months later
- **Security:** Built for secure facilities, not the cloud
CPU-only inference, **zero internet** :
it runs entirely inside an **air-gapped lab**, with compact int8 models



TECHNOLOGY STACK



Python 3.10+

Scientific computing ecosystem



PyTorch + TorchScript

Train models, export compact int8 versions for CPU



FastAPI

Serves the AI to the web console



rasterio

Reads satellite GeoTIFF images with map coordinates



React 18 + TypeScript + Tailwind

Console interface, type-safe and clean



Leaflet

Draws answer maps and region overlays



SQLite + Docker

Offline single-file storage; runs fully air-gapped on CPU



96 automated tests

Every feature verified before demo

HOW IT WORKS — THE ANVESHA PIPELINE

1



Upload

Analyst gives satellite images + a question in plain English



2



Validate

System checks format, map coordinates, and image pairing



3



Understand

Agent identifies what the question is asking



4



Route to 7 Specialist AI models

Shared ResNet-18 vision backbone (11M parameters, trained on EuroSAT) feeds dedicated models for VQA, captioning, change detection, and optical-SAR fusion



5



Answer with Evidence

Confidence score, highlighted map overlays, exportable GeoTIFF masks



6



Auditable Report

Every AI decision recorded in a step-by-step execution trace (PDF/JSON); outputs include change maps and quantified impact

Air-gapped · No internet · Runs on CPU for ISRO labs

PROVEN TODAY ✓



Working system, not a concept

7 specialist models trained;
live demo runs end-to-end



Measured accuracy

98.9% scene classification ·
0.82 F1 change detection



Runs on commodity CPU

int8 models lose only 0.16%
accuracy



Engineered for reliability

96 automated tests guard
every release

HONEST CHALLENGES ⚠️



Precise object localization
below target



Counting objects is
inherently hard



Domain shift to
real ISRO sensors



CPU speed on very
large scenes

OUR STRATEGY ✓



Interpretable fallback

physics-based indices ship today;
detection-grade upgrade pre-gated



Transparent confidence

every answer labeled with its source;
low confidence → ask, never guess



Band-adaptive models

any sensor auto-detected;
uncertainty safeguard switches to
physics-based mode



Tiled int8 inference

large scenes split into tiles;
every run timed and traced



Failed experiments are documented, not hidden — that is our quality guarantee.

IMPACT AND BENEFITS



ANVESHA enables timely decisions, strengthens resilience, and drives sustainable development through accessible, accurate, and actionable Earth Observation intelligence.

github Repository Link:

<https://github.com/Raghavapranav3443/Anvesha>

Foundational Research Papers

- ResNet (scene encoder backbone)

CVPR 2016

<https://arxiv.org/abs/1512.03385>

- Feature Pyramid Networks (FPN) (change detection decoder)

CVPR 2017

<https://arxiv.org/abs/1612.03144>

- Transformer (caption decoder)

NeurIPS 2017

<https://arxiv.org/abs/1706.03762>

- RSVQA (visual question answering on satellite images)

IEEE TGRS 2020

<https://arxiv.org/abs/2003.07333>

Benchmark Datasets Used for Training and Evaluation

- EuroSAT (land cover classification, Sentinel-2)

Dataset: <https://zenodo.org/records/7711810>

Paper: <https://arxiv.org/abs/1709.00029>

- RSVQA-LR (low-resolution VQA triplets)

Dataset: <https://zenodo.org/records/6344333>

- LEVIR-CD (bi-temporal building change detection)

<https://justchenhao.github.io/LEVIR/>

- CDVQA (change detection VQA)

Repository: <https://github.com/YZHJessica/CDVQA>

Paper: <https://arxiv.org/abs/2112.06343>

- BigEarthNet v2 / reBEN (Sentinel-1 + Sentinel-2 pairs for optical-SAR fusion)

Dataset: <https://zenodo.org/records/10891137>

Paper: <https://arxiv.org/abs/2407.03653>

- BigEarthNet.txt (captions and reference boxes for captioning)

<https://huggingface.co/datasets/BIFOLD-BigEarthNetv2-0/BigEarthNet.txt>